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Black spruce (*Picea mariana*) restoration in *Kalmia* heath by scarification and microsite mulching

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ABSTRACT

Canopy removing disturbance such as clearcutting and fire in nutrient-poor boreal and temperate forests often create shrub-dominated vegetation causing significant delay in forest succession. Lack of favourable microsites for black spruce seedling establishment and growth has been identified as the primary cause for conifer regeneration failure in post-fire *Kalmia* heath in eastern Canada. The aim of this study was to compare the growth performance of black spruce planted in scarified plots and in microsites developed by a novel technique called micro-site mulching (MSM) in *Kalmia* heaths. We hypothesized that because of vegetation removal and exposure of mineral soil in scarified and MSM treatments, planted black spruce would suffer from frost heaving and animal browsing, but these treatments may offer better seedling growth by ameliorating the inhibitory effects of *Kalmia* organic matter and providing with favourable soil moisture, soil temperature, soil respiration and nutrient availability. We conducted this study in a 13 year-old post-fire *Kalmia* heath in Triton Brook, Newfoundland, Canada (48°30′N, 50°00′W). We measured black spruce seedling survival and growth annually for three complete growing seasons and soil nutrient availability one year post-planting. In all three treatments black spruce seedling survival was >90%. About 20% of the seedlings in scarified and MSM treatments suffered herbivore damage and frost heaving with little damage in control plots. Black spruce in scarified and MSM plots were associated with significantly higher growth than control, with differences amplified each consecutive year since planting. Higher seedling growth in scarified and MSM plots was associated with higher soil moisture, lower organic matter content, and lower soil temperature. Soil nutrient differences in treatments had little effect on seedling growth. Since MSM treatment produced substantially less soil disturbance with similar or better seedling growth than scarification, we suggest MSM as a preferred site preparation technique in *Kalmia* heath in contexts where large-scale soil disturbance by standard mechanized scarification is undesirable. The novelty of his paper is the development of MSM as a potential alternative to scarification in creating suitable microsites for planted black spruce to restore forest cover in ericaceous heaths. However, an operationally suitable MSM tool must be developed and a longer-term (10 years) monitoring is necessary before the method can be applied as a restoration tool.

1. Introduction

Canopy removing disturbance such as wildfires, clearcutting and extensive insect defoliation is often followed by proliferation of understory plants causing successional delay and interference with forest regeneration (Royo and Carson, 2006). Regaining forest cover after shrub invasion has been a priority for land managers, ecologists and conservation biologists in many parts of the world. In boreal and temperate conifer forests such disturbances have been reported to cause long successional delays due to rapid and prolific vegetative growth of understory ericaceous plants such as *Kalmia angustifolia* L. (Mallik, 1993, 1995; Mallik and Roberts, 1994; Yamasaki et al., 1998; Thiffault et al., 2004a, 2004b), *Calluna vulgaris* L. (Weatherell, 1953; Leyton, 1955), *Vaccinium myrtillus* L. (Mallik and Pellissier, 2000), *Rhododendron groenlandicum* (Oeder) Kron & Judd. (Inderjit and Mallik, 1997) and *Gaultheria shallon* Pursh (Fraser et al., 1995). These ericaceous shrubs resist tree invasion by competition and allelopathy (Mallik, 1987; Zhu and Mallik, 1994), which causes long delay in heath-to-forest transition (Mallik, 2003). The worst examples of successional delay can be found in black spruce (*Picea mariana*)-*Kalmia* communities of eastern Canada, where the ericaceous dominance may last for up to 60 years (Bloom and Mallik, 2006).

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Once *Kalmia* occupies the area, it can modify soil conditions by accumulating partially decomposed litter and by releasing allelochemicals (Inderjit and Mallik, 1999), which increase soil acidity, reduce microbial activity and inhibits organic matter (OM) decomposition and nutrient availability (Mallik, 1994; Bradley et al., 1997). *Kalmia* also interferes with ectomycorrhizal infection of black spruce (Yamasaki et al., 1998). As a result, black spruce recruitment and growth in *Kalmia* dominated sites is extremely poor (Mallik, 2003). Even when seeds are abundant, black spruce seed germination and seedling establishment are limited to areas with thin organic matter (Mallik et al., 2010; Siegwart-Collier and Mallik, 2010). In addition to poor germination, most black spruce seedlings suffer from stunted growth, which further inhibits forest recovery (Mallik and Kravchenko, submitted).

Previous research has shown that in attempts to avoid the seed germination stage by planting young black spruce in *Kalmia* heath, the planted seedlings may suffer from stunted growth (Thiffault and Jobidon, 2006). Removal of *Kalmia* with herbicides provides little benefit to tree seedling growth (Mallik and Inderjit, 2001), because of the inhibitory residual effects of *Kalmia* humus remaining in the soil (Wallstedt et al., 2002; LeBel et al., 2008). Prescribed burning to create favourable seedbeds for conifers requires high temperature burns (Mallik and Roberts, 1994). Application of high temperature prescribed burning is not only expensive and logistically challenging, but also potentially dangerous. Attempts to enhance black spruce growth by fertilizer application showed that the benefits are short lived (LeBel et al., 2008) and have a potential to increase *Kalmia* growth (Mallik, 1996). Mycorrhizal inoculation produced little positive effect on seedling growth, at least within two years after out-planting (Walker and Mallik, 2009; Löf et al., 2012).

To date, a common forest management practice to deal with poor conifer regeneration in *Kalmia* heath has been scarification. This mechanical site preparation technique exposes mineral soil providing substrate favourable for black spruce seed germination (Thiffault et al., 2005). Scarification, planting, and spot fertilization have been reported to increase black spruce growth compared to unscarified treatments (Thiffault and Jobidon, 2006). However, the long-term effects of scarification followed by spot fertilization to maintain high seedling growth are uncertain and cannot be used in conservation areas such as national parks and ecological reserves because of high ground disturbance. An alternative method, suggested by Mallik (1991) and Walker and Mallik (2009) involves mixing organic matter with part of the mineral soil. This mulching technique prevents *Kalmia* regrowth by killing vegetative buds on rhizomes (Mallik, 1991) and improves soil respiration by enhanced decomposition (Walker and Mallik, 2009). But the standard mechanized scarification (Thiffault et al., 2004a) and plot level-soil mulching (Walker and Mallik, 2009) cause high levels of soil disturbance leading to increased erosion, fluctuation of soil moisture and temperature (Thiffault et al., 2004b). Large-scale soil mulching is also expensive and therefore, cannot be used as a routine site preparation method. In a previous study Thiffault et al. (2004a) compared the effect of a very small-scale localized scarification technique (also termed Manual) by using a motorized auger and standard mechanized site preparation treatments (Wadell and harrowing) on planted black spruce seedlings. Although they found the initial seedling growth to be similar across the treatments, 10-y results were poor in the localized (Manual) treatment. It is quite likely that the area mulched by auger (<15 cm diameter) in the localized treatment was too small to create a favourable competition-free microsites for the planted seedling. We thought that if soil mulching is done at a microsite level, 0.8–1 m diameter plots (hereafter referred to as microsite soil mulching or MSM), it might alleviate some of the problems associated with large-scale site preparation by scarification and mulching, while retaining their positive effects of improved microsite conditions.

However, two potentially negative effects might be associated with scarification and MSM technique, frost heaving and animal browsing. The mineral soil exposed scarified sites in the cold, wet, organic-rich boreal forests may experience frost heaving, a phenomenon caused by rapid fluctuation of day and night temperature specially in early spring that often push the planted seedlings out on the surface. Secondly, seedlings planted in the MSM sites free from ground vegetation, can be visible to browsing animals (Miller et al., 2006). Furthermore, fresh growth of planted seedlings in the MSM and scarified sites would be generally nutrient-rich and animals are easily attracted to eat these soft and nutrient-rich tissue (Hjalten et al., 1993). Thus, both frost heaving and animal browsing can affect seedling survival and growth in the scarified and MSM plots.

The objective of this study was to compare scarification and MSM techniques in terms of their capacity to create favourable microsites for planted black spruce in post-fire *Kalmia* heath. We hypothesized that because of vegetation removal and exposure of mineral soil in scarified and MSM treatments, planted black spruce would suffer from frost heaving and animal browsing, but these treatments may offer better seedling growth by ameliorating the inhibitory effects of *Kalmia* organic matter and with favourable soil moisture, soil temperature, soil respiration and nutrient availability.

2. Methods

2.1. Study area

We conducted this study in a 13 year-old post-fire site (Triton Brook) in Newfoundland (48°30′N, 50°00′W) in the Central Newfoundland ecoregion of the Boreal Shield ecozone. Climate in the region is cool and moist where average January temperatures vary from −14° to −10 °C, average July temperatures vary between 11 °C and 15 °C and the area receives average annual precipitation between 801 and 1200 mm, of which approximately 30% is snowfall (Power, 2000). Growing season length is from early June to end of August. The soils of the area are nutrient poor podzolic gravel–loam (Soil Classification Working Group, 1998). Organic matter depth at the study area is very variable ranging from 6 to 34 cm. Mineral soil is composed of large rocks embedded in silty sand surficial till (10–20 cm). Micro-topography was uneven with typical ericaceous hollows and hummocks (Gimingham, 1972) some time rising as much as 40 cm. The post-fire heath was dominated by *Kalmia*, *Rhododendron canadense* [(L.), Torr], *R. groenlandicum* [Oeder (Kron and Judd)] and *Vaccinium* spp., scattered black spruce, tamarack (*Larix laricina* (Koch) seedlings and saplings and specked alder, Alnus *incana* (L.) Moench subsp. *rugosa* (Du Roi) R.T. Clausen. Ground cover was dominated by lichens such as *Cladonia borealis* (Stenroos) and *Cladina rangiferina* [Aiton (Raf.)]. Mean ericaceous cover in the study sites was 74.1 ± 7.2% with *Kalmia* cover being 51.8 ± 6.8% (measured from 15 1 × 1 m random quadrats).

2.2. Experimental design

We established three 100 × 100 m blocks in random locations within a 5 ha study site. Each block was divided into four 25 × 100 m plots (Fig. 1) and randomly assigned to one of the following treatments: (1) unplanted control plots (used for another study), (2) black spruce planted in *Kalmia* (experimental control), (3) black spruce planted after microsite mulching, and (4) black spruce planted after scarification with T26. The site preparation treatments (scarification and MSM) were applied in mid-August,

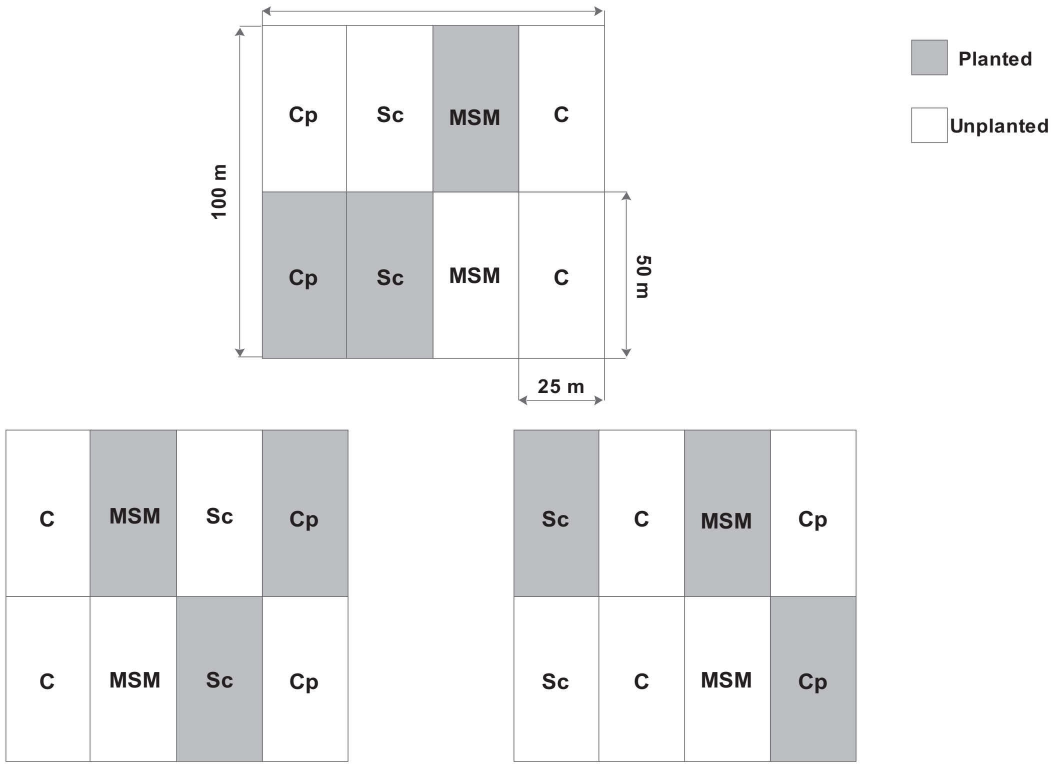

Fig. 1. Experimental design showing scarification, MSM and control treatments in three study blocks. The letters within the rectangles correspond to treatments: control (C), control planted (Cp), scarified (Sc) and microsite soil mixing (MSM). Shaded parts represent areas planted with black spruce. Clear portions represent the parts left unplanted. Unplanted control (C) plots were used for another study (Mallik and Kravchenko, submitted) to assess natural regeneration of black spruce. Therefore, control planted (Cp) was the experimental control for this study and scarification (Sc), microsite soil mixing (MSM) were the two site preparation treatments.

2008 and were followed by planting half of each plot with 150 commercially grown multi-pot containerized black spruce seedlings raised from local seed source (Gander, central Newfoundland) at Woodale Tree Nursery, Newfoundland. Initial mean height and basal diameter of 15 pre-planted seedlings were 23.0 ± 2.0 cm and 2.3 ± 0.3 mm respectively. The other half of the plot was left unplanted to be used for a different study. The unplanted control plots were used to assess natural regeneration of black spruce in *Kalmia* heath for another study (Mallik and Kravchenko, submitted).

The scarification treatment was applied using a disc scarifier (T26, BRACKE FOREST, Bräcke, Sweden) attached to a skidder. This machine works by cutting the organic layer and flipping it aside. The treatment resulted in long parallel trenches of exposed mineral soil about 60 cm wide and positioned 3 m apart. During planting, black spruce seedlings were placed on the hinge every 3 m as per commercial tree planting protocol. MSM treatment was applied by removing vegetation in about 1 m diameter circular area using a brush saw followed by mixing organic and mineral soil horizons in the middle of the plot (0.8 m diameter) with a portable roto-tiller. The result was a relatively undisturbed area with approximately 0.8–1 m diameter mulched microsites in a grid, with 3 m intervals between neighbouring spruce seedlings. One black spruce seedling was planted at the centre of each mulched microsite. Black spruce seedlings planted at the same spacing in undisturbed *Kalmia* heath served as experimental control.

We randomly established five 20 × 20 m permanent sampling quadrats in each 25 × 50 m sub-plot. In each of these plots three 1 × 1 m seedling-centred quadrats were marked and measured for seedling growth and microsite environmental parameters (see below). In total 60 1 × 1 m seedling-centred quadrats were measured for seedling growth response and environmental parameters per treatment block. Immediately after planting in August 2008, height and basal diameter of the seedlings were recorded for all treatments. Black spruce response was determined as the difference in height and basal diameter of seedlings between early June and mid-August in 2009 (year 1), 2010 (year 2) and 2011 (year 3). For environmental parameters we measured soil moisture, soil temperature, soil respiration, and organic matter depth at 10 cm from the black spruce stem in each 1 × 1 m seedling-centred quadrat. Because of budget constraints we were not able to measure foliar nutrients of the planted seedlings in these treatments.

We measured soil temperature and soil moisture from 11:00 to 15:00, preceded by at least 48 h of no precipitation using HH2 Moisture meter and WET-2 sensor (Delta-T Devices, Cambridge, UK). The soil respiration in these plots was measured by using EGM-4 meter with SR-1 probe (PP systems, MA, USA). Organic layer depth was determined by examining two 5 cm diameter soil cores taken in each quadrat 10 cm from the selected black spruce stems that were measured for growth. Herbivore and frost heaving damage of seedlings was assessed visually by inspecting all the planted seedlings assigning a value of 0 (no damage) or 1 (damaged).

We measured soil nutrient availability using plant root simulator (PRS™) probes (Western Ag Innovations Inc., Saskatoon, SK, Canada) in July 2009 (year 1). In each study plot, eight sets of probes, forming two samples, were installed for two weeks in randomly selected quadrats. The PRS probes work by binding positively and negatively charged ions to an ion-exchange resin membrane (Huang and Schoenau, 1997; Hangs et al., 2004; Adesemoye1 and Kloepper, 2008). The values were expressed as amount of nutrients absorbed by 10 cm^2 membrane over a period 14 days. The PRS probes were placed in soil vertically keeping the upper side of 5 cm long membranes just below the surface. Each set of PRS consisted of two probes (anion and cation probes), which were installed next to each other. Immediately after extraction from the soil, the PRS™ probes were washed with distilled water then put in plastic bags for transportation. In the laboratory the probes were cleaned of any residual soil, placed in clean plastic bags, and kept cool, until sent for analysis to Western Ag Innovations Inc., Saskatoon, SK, Canada. For chemical analysis, the four sets of probes were combined to make one sample. The ions bound to the membranes were extracted and their concentration determined. For further details on extraction methods and chemical analyses see Hangs et al. (2004).

2.3. Statistical analyses

Prior to analysis all the data were checked for normality using the Shapiro–Wilks test. Deviations were corrected using $\log(x)$ transformation. Growth of black spruce seedlings in scarified, MSM and control treatments were compared according to a nested design and with a repeated-measures approach. The treatment sub-plots were nested within the blocks. By doing so it was possible to account for variation in data associated with differences among the blocks, and to achieve increased precision of the analyses. A repeated-measures element was introduced to account for multiple measurements of black spruce height, basal diameter, and microsite conditions taken in the same quadrats over a course of several years.

The overall comparison of black spruce growth and size was made by mixed linear model (R function: lme; library: nlme; R Development Core Team, 2010). It allowed the data analysis with levels of hierarchy accounting for variations such as location of the study blocks. We used the mean basal diameter and height as well as basal diameter and height increments in each sub-plot as the response variables. Treatment type (scarification, MSM, control) was considered as a fixed parameter, block as random intercept, and year as random slope. Seedling size and growth differences among the treatments in individual years were analysed by constructing separate mixed linear models. In the models, mean values per sub-plot of seedling height, basal diameter, height increments or basal diameter increments were used as the response variables. Treatment type (scarification, MSM, control) was considered as a fixed parameter and block as random intercept. Although a mixed model ANOVA, incorporating time since planting, would have been ideal for this study design this approach was not possible due to high damage and mortality rates of seedlings. It was not possible to repeatedly sample all seedlings. To account for the large imbalance in the sample sizes within treatments among years we were forced to analyse and interpret time periods separately.

To determine the relationship between black spruce growth and microsite conditions, further analyses were conducted using growth and environmental (soil moisture, soil temperature, soil respiration, and organic matter depth) data for individual 1×1 m quadrats. The effects of herbivore damage and frost heaving on seedling survival in year one were estimated using multiple regressions. For this analysis, survival of all seedlings after one year was used as a response variable, while presence/absence of herbivore and heaving damage were used as independent factors. Environmental factors associated with higher growth were identified using step-wise multiple regressions (R functions: lm, step; library: stats, R Development Core Team, 2010). The data were checked for correlations. All measured parameters were well within the set range (0.70 > Pearson correlation coefficient < −0.70) and were included in the preliminary model. Because seedling height increments were affected by browsing basal diameter increments for each sampled seedling were used as response, while environmental measurements in each plot as independent variables.

To account for growth differences among black spruce in the three treatments, biophysical and soil nutrient conditions were plotted using non-metric multidimensional scaling (NMDS) ordination (R function: metaMDS; library: vegan, R Development Core Team, 2010). The points were plotted based on environmental (soil moisture, soil temperature, soil respiration, and organic matter depth) or nutrient data in corresponding quadrats. The relationship between seedling growth and microsite conditions was determined as a correlation between basal diameter increments in each quadrat and corresponding microsite conditions (function: envfit; library: vegan, R Development Core Team, 2010). The results were plotted on NMDS plot. The relationship between seedling basal diameter increments and microsite coordinates in ordination space were expressed as an arrow indicating a position in ordination-diagram where black spruce growth was the highest among the microsites. The length of an arrow corresponded to the magnitude of variation in seedling growth relative to the position of the points.

3. Results

3.1. Black spruce seedling survival

Survival of planted black spruce seedlings in all treatments was above 90% (Fig. 2). There was no significant difference in seedling survival rates among the treatments (Table 1). Planted black spruce seedlings in MSM microsites experienced a higher degree of herbivore damage (likely from snowshoe hare) compared to control, with 22.2% seedlings showing signs of browsing (Fig. 2). Seedlings in the scarified treatment were less browsed (11.3%) than those in MSM treatment. Black spruce seedlings in scarification treatment were susceptible to frost heaving (10.2%) where large portions of the root system were pushed out of soil. Black spruce seedlings, in control microsites experienced very little browsing (3.5%) and no heaving damage. Neither browsing ($\beta = 0.01$, Sig.$T = 0.65$) nor heaving ($\beta = -0.04$, Sig.$T = 0.105$) had a significant effect

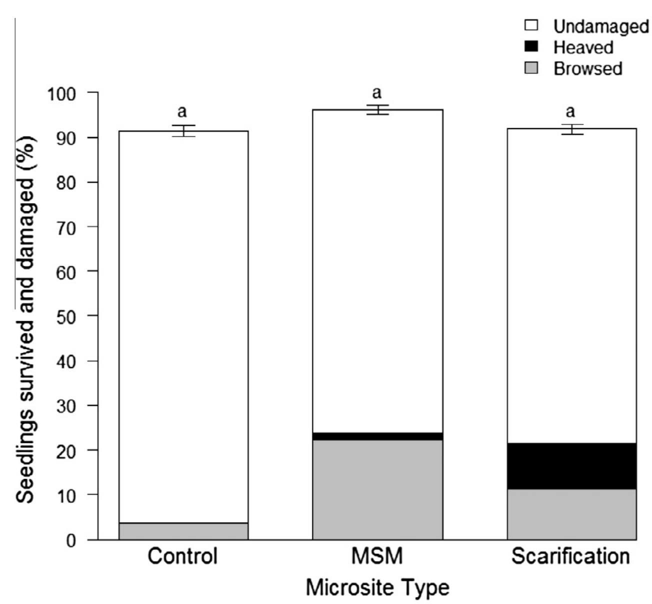

Fig. 2. Black spruce seedling survival and proportion damaged by browsing and frost heaving in control (Cp), microsite soil mixing (MSM) and scarification (Sc) treatments in year 1. Total column height represent percent of seedlings survived in each treatment. Shaded areas are percent of seedlings damaged by herbivores or frost heaving. Same letter indicates no significant difference ($p = 0.05$).

Table 1
Results of mixed linear regression on the planted black spruce seedling survival in scarified, MSM and control plots in year 1. Values for $p < 0.05$ indicate significant difference from control.

Model	Comparison	β	SE β	p
Seedling survival in year 1	Scarification control	0.001	0.052	0.98
	MSM control	0.02	0.052	0.67
	Scarification MSM	0.02	0.052	0.69
		Log Lik = 6.34	DF = 4	

($r^2 = 0.0021$, $F_{2,1349} = 1.43$, $p = 0.24$) on the survival of black spruce seedlings two years after treatment.

3.2. Black spruce seedling growth

At the time of planting, seedling height did not vary significantly (Table 2) among the treatments but had a slightly higher basal diameter in the MSM treatment. In years 1–3, black spruce basal diameter and height became noticeably greater in scarification and MSM than control (Fig. 3A and B). Seedling height was significantly higher in scarified and MSM plots compared to control. Although in year 1 seedling growth did not vary significantly among the three treatments (Table 3) in years 2–3 both height and basal diameter increments were considerably greater in scarification and MSM treatments than in control (Fig. 3C and D).

3.3. Microsite conditions

In year 2, black spruce seedling growth had a significant relationship with soil moisture and temperature (Table 4). Step-wise multiple regression analysis explained 38% of the variance in black spruce basal diameter increments ($r^2 = 0.41$, $r^2_{adj} = 0.38$). The analysis revealed a direct positive relationship of black spruce basal diameter increments with soil moisture and inverse relationship with soil temperature.

In year 3, the 34% variation in black spruce basal diameter increment ($r^2 = 0.37$, $r^2_{adj} = 0.34$) was explained by a combined effect of organic matter (OM) depth and soil respiration (Table 4). Unlike the previous year, OM depth played a greater role in black spruce growth with a significant negative effect. Variation in microsite conditions along axis two of the NMDS ordination was associated with increasing soil moisture and decreasing soil temperature (Fig. 4). Seedling growth was correlated with low OM depth and high soil moisture. Variation in microsite conditions in year 2 along axis 2 of the NMDS ordination was associated with seedling growth (basal diameter), increased soil moisture and decreased soil temperature (Fig. 4). The highest seedling growth was found in quadrats located in top right corner of the ordination space, which were characterized by relatively high soil moisture, low soil temperature and shallow OM depth. These were mostly MSM plots (Fig. 4).

Ordination analysis of year 3 environmental data also produced a two dimensional solution accounting for 98% (stress = 14.0) of cumulative variance of environmental parameters measured in each quadrat. As in year 2, OM gradient was associated with axis one, with higher values towards the negative end of the axis (Fig. 5). Seedling growth was similar to year 2 but more correlated with higher temperature. Distribution of points along the first axis corresponded to treatment type. Scarification, MSM, and control microsites formed three closely located clusters in the ordination space. The scarified plots were concentrated towards the right of the ordination diagram the control plots were in the left while the MSM plots were in the middle (Fig. 5).

Available soil nutrient data showed that scarification, MSM and control plots were not that different in terms of available N but P and K availability was the highest in MSM treatment (Table 5). On the other hand Fe and Al availability was the highest in the scarified treatment. The NMDS ordination produced a two dimensional solution explaining 99% (stress = 10.4) of cumulative variance in available nutrients. Although the three treatment plots

Table 2
Results of multiple mixed linear regressions of the planted black spruce seedling's basal diameter and height in scarified, MSM and control plots. Values for $p < 0.05$ represent indicate difference from control.

Model	Comparison	Basal diameter			Height		
		β	SE β	p	β	SE β	p
Years 0–3 combined	Scarification control	0.76	0.17	<0.001	2.26	1.01	0.002
	MSM control	0.83	0.17	<0.001	3.54	1.01	0.04
	Scarification MSM	0.07	0.17	0.68	1.28	1.01	0.22
		Log Lik = −0.04	DF = 15		Log Lik = 1.68	DF = 15	
Year 0	Scarification control	0.1	0.13	0.50	−0.94	1.73	0.60
	MSM control	0.2	0.13	0.20	1.12	1.73	0.50
	Scarification MSM	−0.1	0.13	0.47	−2.06	1.73	0.29
		Log Lik = −0.34	DF = 4		Log Lik = −15.46	DF = 4	
Year 1	Scarification control	0.23	0.06	0.02	−0.03	0.72	0.90
	MSM control	0.41	0.06	0.002	1.68	0.72	0.08
	Scarification MSM	−0.18	0.06	0.04	−1.71	0.72	0.07
		Log Lik = 4.66	DF = 4		Log Lik = −10.35	DF = 4	
Year 2	Scarification control	0.59	0.09	0.002	2.49	1.44	0.16
	MSM control	0.7	0.09	0.001	4.57	1.44	0.03
	Scarification MSM	−0.11	0.09	0.26	−2.07	1.44	0.22
		Log Lik = 1.07	DF = 4		Log Lik = −13.57	DF = 4	
Year 3	Scarification control	1.7	0.23	0.01	5.12	2.08	0.09
	MSM control	1.38	0.23	0.01	4.36	2.08	0.10
	Scarification MSM	0.32	0.23	0.24	0.76	2.08	0.74
		Log Lik = −2.66	DF = 4		Log Lik = 2.28	DF = 4	

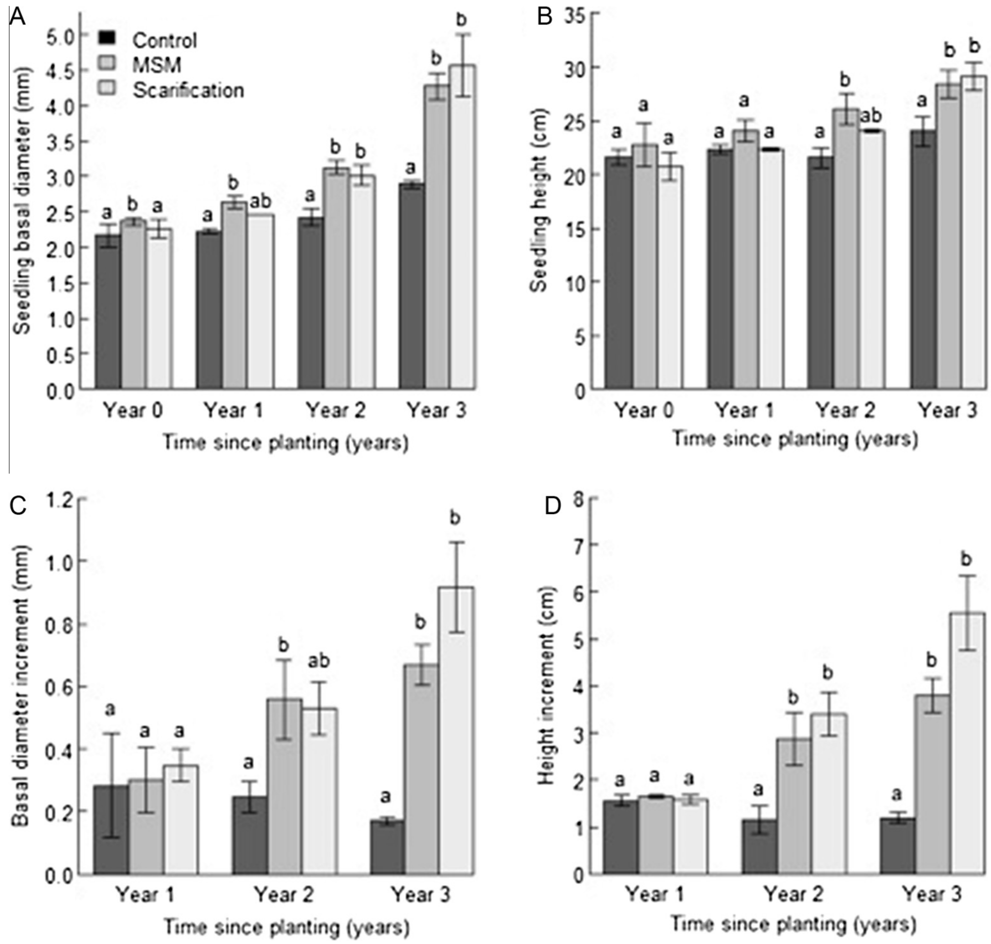

Fig. 3. Variation in mean (A) basal diameter, (B) height, (C) basal diameter increments (early June – late August) and (D) height increments of seedlings in control (Cp), microsite soil mixing (MSM) and Scarified (Sc) plots 1–3 years after treatment. Panels in Fig. 3A and B show respectively height and basal diameter of seedlings planted at the end of the growing (August 2008) referred to as year 0. Treatment response was monitored at the elapse of three full growing seasons (2009–2011). Unlike letter(s) above the histograms represent significant difference among treatments with in the same year.

Table 3
Results of multiple mixed linear regressions on planted black spruce basal diameter increments and height increments in scarified, MSM and control plots. Values for $p < 0.05$ indicate significant difference from control.

Model	Comparison	Basal diameter increments			Height increments		
		β	SE β	p	β	SE β	p
Years 1–3 combined	Scarification control	0.36	0.09	0.001	2.21	0.53	<0.001
	MSM control	0.28	0.09	0.01	1.47	0.53	0.01
	Scarification MSM	0.09	0.09	0.35	0.74	0.53	0.17
		Log Lik = −1.49	DF = 15		Log Lik = −42.77	DF = 15	
Year 1	Scarification control	0.06	0.12	0.6	0.02	0.11	0.9
	MSM control	0.02	0.12	0.9	0.08	0.11	0.5
	Scarification MSM	0.04	0.12	0.72	−0.06	0.11	0.61
		Log Lik = −0.03	DF = 4		Log Lik = 1.68	DF = 4	
Year 2	Scarification control	0.28	0.11	0.06	2.25	0.61	0.02
	MSM control	0.31	0.11	0.05	1.72	0.61	0.05
	Scarification MSM	−0.03	0.11	0.81	0.52	0.61	0.43
		Log Lik = 1.02	DF = 4		Log Lik = −8.69	DF = 4	
Year 3	Scarification control	0.75	0.10	0.001	4.364	0.73	0.004
	MSM control	0.5	0.10	0.01	2.6	0.73	0.02
	Scarification MSM	0.25	0.10	0.06	1.76	0.73	0.07
		Log Lik = 1.42	DF = 4		Log Lik = −9.46	DF = 4	

Table 4
Results of step-wise multiple regression analyses on planted black spruce seedling basal diameter increments as a function of microsite conditions in year 2 and 3.

Model	Partial regression coefficients					
	Independent variables	Intercept	*b*	SE *b*	*β*	Sig.*T*
Year 2	Soil moisture	0.17	0.002	0.001	0.35	0.02
	Soil temperature	0.17	−0.01	0.002	−0.39	0.01
	Overall model fit	$r^2 = 0.41$	$F = 14.48$	DF = 2, 42	$p < 0.001$	
Year 3	OM depth	0.98	−0.03	0.01	−0.46	0.002
	Soil respiration	0.98	−1.22	0.63	−0.26	0.056
	Overall model fit	$r^2 = 0.37$	$F = 12.52$	DF = 2, 42	$p < 0.001$	

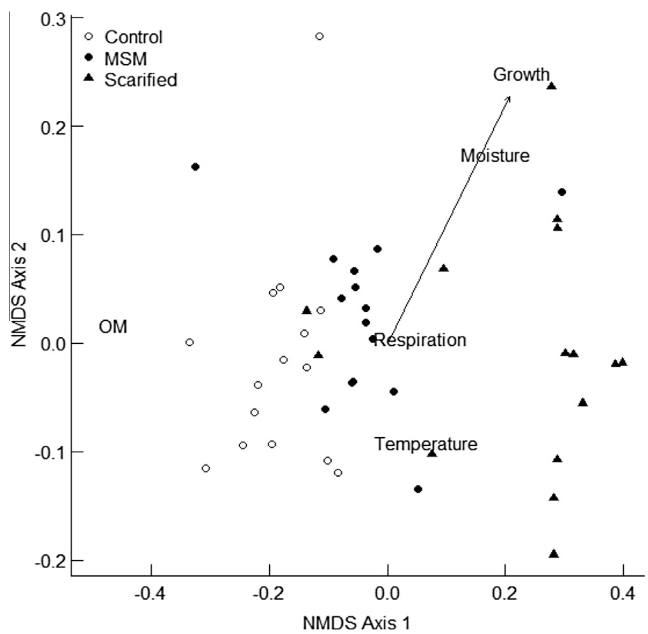

Fig. 4. NMDS ordination showing seedling growth in relation to microsite environmental conditions among control (Cp), microsite soil mixing (MSM) and scarification (Sc) treatments in year 2. Points represent individual sampling quadrats in the three treatments. The point coordinates were determined based on microsite environmental conditions in year 2. Texts in the ordination diagram indicate distribution of individual environmental parameters along the axis. Arrow corresponds to variation in seedling basal diameter (growth) increments in relation to position of points in ordination space ($r = 0.64$). Seedling growth was correlated with lower organic matter (OM) and higher soil moisture.

did not produce distinct clusters, their distribution was limited to certain areas on the ordination plot (Fig. 6). Control plots were aggregated at the bottom-left and were associated with higher content of metals such as Mn, Ca, Zn and Mg. Distribution of MSM plots was mostly limited to the top-left portion of the diagram. These were characterized by high available P. Most of the scarified plots were located on the right side of the ordination space and were associated with high available Fe.

Seedling basal diameter increments fitted to the ordination plot indicated that the diameter growth rates were higher in scarified and MSM plots than in control (Fig. 6). However, a relatively low correlation coefficient ($r = 0.34$) suggested that the difference in seedling growth was not directly associated with the variation in available nutrient conditions.

4. Discussion

We hypothesized that scarification and MSM may cause damage to planted seedlings but the treatments would enhance black spruce seedling growth from improved microsite environmental conditions. Our results showed that seedling survival in scarified, MSM, and control plots was above 90% with no significant difference among the treatments. However, black spruce seedlings in scarified and MSM treatments were more susceptible to damage from herbivores and frost heaving than control. On the other hand, height and basal diameter increments in scarified and MSM treatments were much greater than control. Thiffault et al. (2004a) and Thiffault and Jobidon (2006) reported positive effects of scarification on black spruce growth. Walker and Mallik (2009) reported increased black spruce seedling growth and higher available soil nutrients in mechanically mulched plots (4.5 × 4 m) compared to control (unmulched) plots. In the current experiment a similar increase in black spruce growth was obtained by MSM treatment, which created significantly less ground disturbance compared to that of Walker and Mallik (2009). The management advantage of

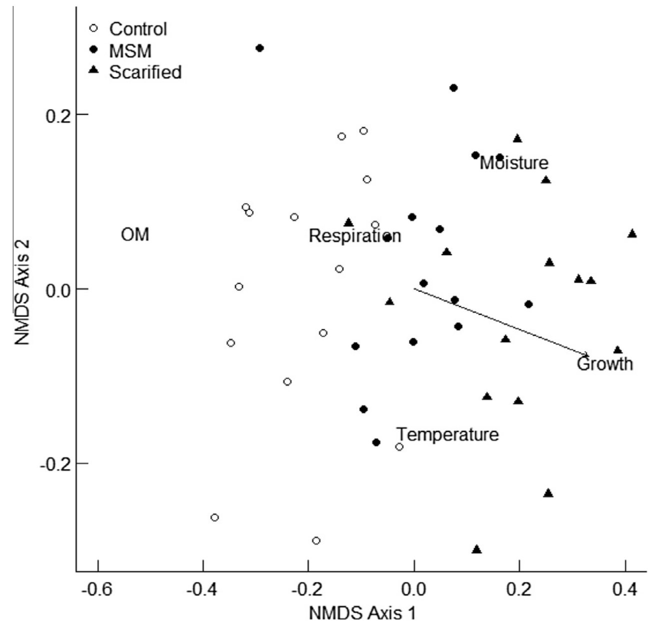

Fig. 5. NMDS ordination showing seedling growth in relation to microsite environmental conditions among control (Cp), microsite soil mixing (MSM) and scarification (Sc) treatments in year 3. Points represent individual sampling quadrats in the three treatments. The point coordinates were determined based on microsite environmental conditions in year 3. Texts in the ordination diagram indicate distribution of individual environmental parameters along the axis. Arrow corresponds to variation in seedlings basal diameter (growth) increments in relation to position of points in ordination space ($r = -0.57$). Seedling growth was similar to year 2 but more correlated with higher soil temperature.

Table 5
Available soil nutrients (mean ± SE) expressed as µg/10 cm^2/14 days (see methods for details) one year after planting black spruce seedling in scarified, MSM and control plots.

Nutrient type	Treatments		
	Control	MSM	Scarification
N (NO_3N & NH_4N)	18.4 ± 2.1	17.5 ± 1.5	18.9 ± 1.9
P	1.6 ± 0.43	3.5 ± 1.47	0.6 ± 0.12
Ca	181.9 ± 25.5	117.8 ± 14.8	110.2 ± 16.7
Mg	82.3 ± 18.3	64.4 ± 7.2	61.0 ± 12.0
K	63.4 ± 7.2	142.4 ± 24.5	100.1 ± 7.36
Fe	2.1 ± 1.1	1.5 ± 0.19	13.7 ± 10.1
Mn	8.9 ± 2.3	5.7 ± 1.64	2.7 ± 0.88
Zn	0.73 ± 0.08	0.6 ± 0.1	0.5 ± 0.11
B	1.03 ± 0.16	0.93 ± 0.16	0.8 ± 0.15
S	41.4 ± 2.5	44.6 ± 3.26	53.2 ± 3.72
Al	12.1 ± 1.46	11.2 ± 0.88	17.1 ± 3.22

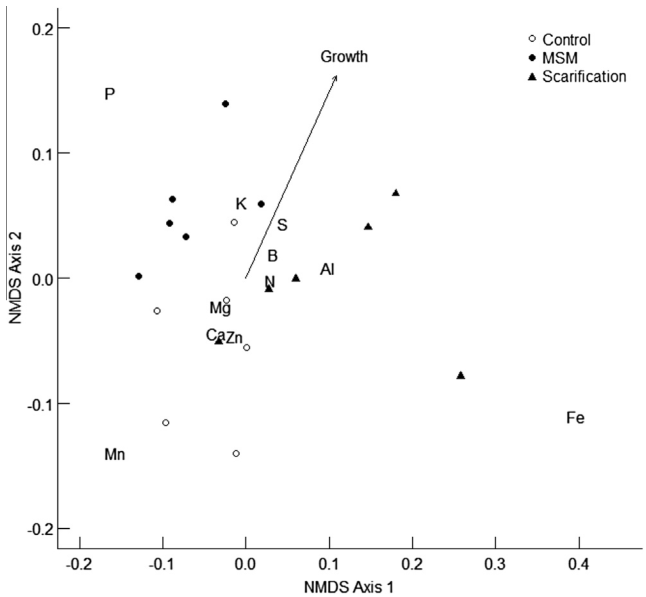

Fig. 6. NMDS ordination showing available soil nutrients measured by ion exchange resin method (PRS probes) in control, MSM and scarification treatments 11 months after the treatments (first growing season, July, 2009). Points represent samples in each of three treatments. Texts in the ordination diagram indicate distribution of individual nutrients along the axis. Arrow corresponds to variation in seedlings basal diameter (growth) increments in relation to position of points in the ordination space ($r = 0.34$).

MSM is its minimal ground disturbance, which would make its use particularly appropriate for ecologically sensitive areas such as national parks, ecological reserves and other conservation areas.

To determine which treatment creates the best conditions for planted seedlings, it is necessary to consider several factors, (i) seedling survival, (ii) risk of damages and (iii) growth. Seedling survivals were similar in all three treatments. Seedlings in control microsites were less likely to be damaged because *Kalmia* cover protects them from herbivore and frost heaving damage but they experienced slow growth. In *Kalmia* heath, black spruce seedlings may remain stunted for decades, causing long delays in forest succession after wildfires (Mallik et al., 2010; Mallik and Kravchenko, submitted). Although seedlings in scarified and MSM plots suffered from animal browsing and frost heaving, they were able to recover quickly and grow at a much higher rate than those in the control plots. Therefore, both of these site preparation techniques were effective in black spruce seedling regeneration. The degree of soil mixing is critical for favourable microsite creation for planted seedlings. Recently, Henneb et al. (2015) reported that for paludified conditions of the Clay Belt region of Canada, forest harrow is a better site preparation equipment than the T26 scarifier presumably for its better soil mixing ability.

In year 2 we found that 38% of the variation in seedling growth to be explained by the combined effects of soil moisture and soil temperature. Soil temperature was inversely associated with seedling growth. In a test of root elongation of six boreal species in response to soil temperature, Tryon and Chapin (1983) found higher root growth of black spruce in colder and wetter sites. This result corresponds to our findings of greater black spruce basal diameter and height increments in microsites with lower soil temperature and higher soil moisture. Only a small number of scarified microsites showed good growth of black spruce (Fig. 6). Thiffault and Jobidon (2006) demonstrated that scarification could result in higher soil moisture compared to control. However, they also found higher soil temperature fluctuation in scarification treatment, particularly in summer. Our ordination results demonstrate that even though scarification produced good conditions for some seedlings, many of the scarified microsites did not provide the optimal growing conditions presumably due to lack of moisture and higher soil temperature. On the other hand, MSM treatment produced consistently better growing conditions for black spruce than undisturbed *Kalmia* heath (control). As a result, mean seedling growth was similar in microsites produced by both scarification and MSM.

In year three 34% of variance in black spruce basal diameter increments was explained by the combined effects of OM depth and soil respiration. Contrary to results from year 2, soil temperature and soil moisture had little or no effect on black spruce seedling growth. Unlike the previous year, OM depth appeared to be a predominant factor in black spruce growth, greater basal diameter increments in microsites with shallow OM. Many studies have demonstrated that *Kalmia* litter, which makes up a large portion of the organic layer in *Kalmia* heath, can have negative effects on black spruce growth (Mallik, 1994, 2001; Bradley et al., 1997; Inderjit and Mallik, 1999; Prescott et al., 2000). Higher decomposition associated with greater soil respiration is thought to alleviate the negative effects of *Kalmia* through degradation of allelochemicals (Zeng and Mallik, 2006) and release of locked up soil nutrients (Staaf and Berg, 1982). It has also been demonstrated that soil mulching can enhance decomposition (Walker and Mallik, 2009) and is associated with elevated soil respiration (Mallik and Hu, 1997). Therefore, we expected that soil respiration would have a positive effect on black spruce growth. However, our data in year 3 revealed an inverse relationship between soil respiration and black spruce basal diameter increments (Fig. 5). This result is likely due to relatively low respiration rates in the scarified plots (Mallik and Hu, 1997) that primarily have a mineral soil substrate. It is also likely that a snapshot of environmental data collected at a selected time did not reflect the overall near-ground environmental conditions of the microsites in the three treatments. Acknowledging this limitation one must be cautious in interpreting the effect of environmental variables on black spruce seedling growth.

Previous studies have suggested that one of the main mechanisms by which *Kalmia* inhibits conifer growth is by controlling soil nutrients (Bradley et al., 1997; Inderjit and Mallik, 1999; Yamasaki et al., 2002). Loose clusters of plots in our NMDS ordination (Fig. 6) suggest that scarification and MSM produce distinctly different microsite conditions. The distribution of points in ordination space did not follow a single gradient, which might be interpreted as a decreasing proportion of organic and mineral soil components. We conclude that axis one is directly related to the mineral soil substrate as evidenced by high concentration of Fe on the right part of the ordination plot (Fig. 6). *Kalmia* has been noted to induce iron pan formation (Damman, 1971). Scarification exposed the mineral soil with broken iron pans creating a substrate with high Fe. Axis two is strongly associated with available P (Fig. 6). Phosphorous concentrations were highest in MSM and intermediate in control, and lowest in scarification treatment (Table 5). This trend closely corresponds with soil respiration values in the three treatments. *Kalmia* has a strong capacity to affect soil nutrient availability (Inderjit and Mallik, 1996, 1999; Bloom and Mallik, 2006). *Kalmia* cover has an inverse relation with total soil N and P concentrations. Organic matter decomposition results in the release of locked up P and N (Staaf and Berg, 1982). Hence, removal of *Kalmia* and enhanced decomposition due to soil mixing may have resulted in higher P concentration in the MSM plots. By the same logic we could expect a similar trend for available N. However, total available N varied only slightly among the treatments. This may be due to very low available N in this organic matter rich *Kalmia* dominated heath (Bloom and Mallik, 2006).

While our ordination results indicate that black spruce seedlings performed better in microsites created by scarification and MSM, the growth rates were not associated with any particular nutrient. This is likely due to low availability of N, which acts as a limiting factor in this system (Bloom and Mallik, 2006). Although the three types of microsites did vary in nutrient conditions, the mechanism by which they affect black spruce growth does not seem to be based on nutrient availability at least one year after the treatments. The inclusion of foliar nutrient data of the planted seedlings would have been useful in explaining their growth response due to treatments. The differences in environmental factors associated with seedling growth between year 2 and 3 confirmed the temporally dynamic and complex nature of the microsites (Munson et al., 1993).

Although this study was conducted in one post-fire *Kalmia*-dominated site in Newfoundland, the site is typical of most boreal and temperate forests on nutrient-poor coarse textured soils with thick organic matter dominated by conifers with ericaceous understory where canopy disturbance by clear cutting and fire results ericaceous dominance, which interferes with forest regeneration (Gimingham, 1972; Mallik, 1993, 1995; Mallik and Roberts, 1994; Fraser et al., 1995; Inderjit and Mallik, 1997; Yamasaki et al., 1998; Mallik and Pellissier, 2000; Thiffault et al., 2004a, 2004b, 2005). Therefore, the results of our study would be widely applicable to many boreal and temperate sites worldwide.

5. Conclusions and management implications

From a forest management perspective it is important to consider not only the ecological effects, but also the logistical problems and public opinion associated with forest management including site preparation treatments (Burton et al., 2003). In some areas, it is acceptable to use commercially available methods, such as scarification, associated with high level of ground disturbance. However, in protected areas, such as national parks and ecological reserves, this may not be a desirable option. With strong pressures from the public to conserve the natural environment and opposition to the use scarification as a site preparation technique that causes significant ground disturbance through increasing soil erosion and sediment transport in streams (John Gosse, Pers. Comm.; Visitor's survey of Terra Nova National Park), land managers are left with very few options.

Our study demonstrates that MSM can enhance black spruce growth similar to scarification in the short-term, but with minimum soil disturbance. If black spruce seedlings were planted at a density required for canopy closure (2600 stems/ha, Newton and Weetman, 1993), the disturbed area from MSM would be one forth (1313 ± 159 m^2/ha, estimated from this study) compared to scarification (4456 ± 1000 m^2/ha). Therefore, MSM should be a preferred site preparation method in environmentally and socially sensitive conservation areas where large-scale disturbance is not desirable. However, an operationally suitable MSM tool must be developed and a longer-term (10 years) monitoring is necessary before this novel approach can be applied as a forest restoration option.

Conflict of interest

The authors declare no conflict interest, financial or otherwise.

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